INSTITUTE OF TECHNOLOGY BLANCHARDSTOWN

Academic term	2017/2018
Year of study	YEAR 2
Semester	SEMESTER ONE
Date of examination	Thurs 4th Jan 2018
Time of examination	3.30pm – 5.30pm

Programme code	Programme title	Module code
BN002	Higher Certificate in Science in Computing in Information Technology	COMP H2026
BN013	Bachelor of Science in Computing In Information Technology	COMP H2026
BN104	Bachelor of Science (Honours) in Computing	COMP H2026
BN120	Bachelor of Science (Honours) in Computing in Digital Forensics & Cyber Security	COMP H2026

Module title	Information Technology Mathematics

Internal Examiner(s)	**Laura Keyes**
External Examiner(s)	**Dr. Tom Lunney,** **Dr. Simon Caton**

Instructions to candidates:

1.	**To ensure that you take the correct examination, please check that the module and programme which you are following is listed in the table above.**
2.	**Question 1 is compulsory. Candidates should attempt ALL parts of Question 1, and any other TWO questions.**
3.	**This paper is worth 100 marks. Question 1 is worth 40 marks. The remaining questions are worth 20 marks each.**
4.	**Remember to mark each number and section clearly and to list the questions which you have attempted in the correct order on the front of you exam answer script.**
5.	**Show all your work**

DO NOT TURN OVER THIS PAGE UNTIL YOU ARE TOLD TO DO SO

Question 1: (Compulsory)

a) Given the following matrices:

$$A=\begin{bmatrix} 2 & 1 & 0 \\ -1 & 2 & -2 \\ 1 & 3 & -1 \end{bmatrix},\ B=\begin{bmatrix} 4 & 6 & 0 \\ -3 & 1 & 10 \end{bmatrix},\ C=\begin{bmatrix} 1 \\ 3 \\ -4 \end{bmatrix}$$

i. What is the size of each of the 3 matrices *A*, *B* and *C*?

ii. Write down the value of the elements b_{22} and c_{21}.

iii. Write down the *transpose* of matrix *C*.

(5 marks)

b) Given $A=\begin{bmatrix} 1 & -1 & 2 \\ 3 & 1 & -5 \\ 1 & 1 & -2 \end{bmatrix}$

i. Find the *cofactors* c_{11}, c_{12} and c_{13}.

ii. Use these cofactors to find the *determinant* of *A*.

(5 marks)

c) Describe, with the aid of a diagram, a *rooted tree* structure.

(5 marks)

d) Draw the graph *G* represented by the following adjacency matrix and write down the degree of each vertex in *G*.

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 1 & 2 & 1 \\ 0 & 2 & 0 & 2 \\ 0 & 1 & 2 & 0 \end{pmatrix}$$

(5 marks)

e) Using the dataset 18, 10, 27, 6, 28, 11, 2, 26, compute the following:

i. Mean

ii. Standard Deviation

(5 marks)

f) Evaluate the following probabilities:

i. Getting a five, if a fair dice is rolled.
ii. If two fair dice are thrown, the probability that the product of the numbers is less than 7

(5 marks)

g) Differentiate the function $y = 3x^2 - x$ from first principles.

(5 marks)

h) Evaluate the following indefinite integrals:

i. $\int \frac{6}{x} dx$

ii. $\int (8x^3 + 3x^2 - 6x + 7) dx$

(5 marks)

Total: 40 marks

Question 2: Differentiation and Integration

a) Find the slope of the tangent to the curve $y = 2x^3 - 3x + 6$ at the point *(2,10)*.

(5 marks)

b) Differentiate the function $f(x) = (3x + 2)(5x - 7)$ using the product rule.

(7 marks)

c) Determine all maximum points, minimum points and points of inflection of $y = 8x^3 + 4x^2 - 6$.

(12 marks)

d) Find the area enclosed by the curve $y = x + 2x^2$ and the x-axis for *x* between 0 and 2.

(6 marks)

Total: 30 Marks

Question 3: Statistics and Probability

a) When performing data analysis why sometimes is a *sample* of the data used and not the *entire population*? Include in your answer a brief description of TWO possible *sampling techniques* that can be used.

(4 marks)

b) The following (Table 3.1) is a distribution of the exam grades of 42 students in a college Maths assessment:

Table 3.1

Exam Score	Number of Students
0-9	1
10-19	1
20-29	2
30-39	5
40-49	7
50-59	5
60-69	8
70-79	6
80-89	4
90-99	3

Calculate the following for the grouped data represented in table 3.1:

i. the **Relative Frequencies** **(2 marks)**
ii. the **Mean** **(4 marks)**
iii. the **Standard Deviation** **(6 marks)**

c) There are six items in a list. In how many ways can three items be chosen:

i. if the order matters? **(3 marks)**
ii. if the order doesn't matter? **(3 marks)**

d) X is a random variable which is *normally distributed* with a mean of 1.98 and a standard deviation of 0.012. What is the probability that a value of X, chosen at random, is greater than 2?

(8 marks)

Total: 30 Marks

Question 4: Matrices, Graphs and Trees

a) Given the following matrices:

$$A=\begin{bmatrix} 1 & 2 & 3 \\ 4 & 1 & 5 \\ 6 & 0 & 2 \end{bmatrix} \qquad B=\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 4 & 9 & 1 \end{bmatrix}$$

i. Calculate AA^T **(3 marks)**

ii. Calculate the inverse of matrix B **(9 marks)**

b) Find a *minimal spanning tree* for the weighted graph in figure 4.1.

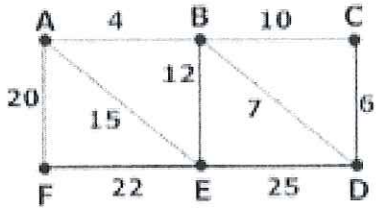


Figure 4.1 **(7 marks)**

c) Find the number of paths of length 2 between *a* and *f* in the graph in figure 4.2 below.

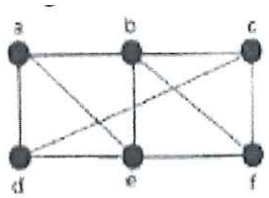


Figure 4.2 **(3 marks)**

d) Describe, with the aid of examples, the difference between *breadth-first* and *depth-first* graph traversals.

(8 marks)

Total: 30 Marks

Formulae

Matrix Determinant

$\det A = ad - bc$ (2x2 matrix)

$\det A = a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13}$ (3x3 matrix using row 1)

Matrix Inverse

$$A^{-1} = \frac{1}{\det A}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$ (2x2 matrix)

$$A^{-1} = \frac{1}{\det A}C^T$$ (3x3 matrix)

General Probability:

If two events A and B are mutually exclusive, then the following law holds:

$$P[A \text{ or } B] = P[A] + P[B]$$

If the two events are not mutually exclusive, the law above is broadened so that the probability of A or B occurring is given by:

$$P[A \text{ or } B] = P[A] + P[B] - P[A \text{ and } B]$$

Consider two events *A* and *B* with event *B* dependent on *A*. The probability of the occurrence of both events is given by:

$$P[A \text{ and } B] = P[A] \times P[B \mid A]$$

Probability Distributions:

$$P[X = r] = {}^{n}C_r p^r q^{n-r} = \frac{n!}{(n-r)!r!} p^r q^{n-r}$$ (Binomial distribution)

$$P[X = r] = \frac{e^{-\mu}\mu^r}{r!}, \text{ where } \mu = np$$ (Poisson distribution)

$$a = \frac{x - \mu}{\sigma}$$ (Normal distribution)

Statistics:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$ (Dataset mean)

$$s = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}} \quad \text{or} \quad s^2 = \frac{\sum_{i=1}^{n} x_i^{\,2} - n(\bar{x})^2}{n-1}$$ (Dataset standard deviation)

$$\bar{x} = \frac{\sum_{i=1}^{n} f_i m_i}{\sum_{i=1}^{n} f_i}$$ (Grouped data mean)

$$s = \sqrt{\frac{\sum_{i=1}^{n} f_i (m_i - \bar{x})^2}{\sum_{i=1}^{n} f_i - 1}} \quad \text{or} \quad s^2 = \frac{\sum_{i=1}^{M} f_i m_i^{\,2} - M(\bar{x})^2}{M-1}$$ (Grouped standard deviation)

Derivation:
First Principles: $\frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$ **Power Rule:** $\frac{dy}{dx} = nx^{n-1}$

Product Rule:

$$\frac{df}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

Quotient Rule:

$$\frac{df}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

Integral: $\int ax^n dx = \frac{ax^{n+1}}{n+1} + c, \; n \neq -1$

Definite Integral:

$$\int_a^b f'(x)dx = f(x)\Big]_a^b = f(b) - f(a)$$

Table of Probabilities – P[Z > a], where Z is the standard normal variable.

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
3.5	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002
3.6	0.0002	0.0002	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.7	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.8	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
4.0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000